

KARTIK TRIPATHI

DATA ANALYST | SQL & BUSINESS INTELLIGENCE DEVELOPER

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PROFESSIONAL SUMMARY

Analytical Data Analyst certified with the IBM Data Analytics Professional Certificate and holding a B.Sc. in Physics, Chemistry, & Mathematics. Rigorous expertise in SQL (Advanced Queries, DDL/DML, Window Functions), Python (Pandas, NumPy, SciPy), Power BI, Tableau, Advanced Excel, and Applied Statistics. Proven ability in architecting normalized relational databases, developing executive BI dashboards, and executing predictive customer RFM segmentations to drive business decision-making.

TECHNICAL SKILLS

- **SQL & Database Engineering:** MySQL, SQLite, PostgreSQL, Relational Schema Normalization (3NF), DDL/DML, Window Functions (ROW_NUMBER), Multi-Table Joins, Subqueries (EXISTS/NOT EXISTS), Set Operations
- **Programming & Analytics:** Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn, SQLAlchemy), Jupyter Notebooks, RFM Customer Segmentation, Pareto (80/20) Decile Modeling, Data Quality Profiling
- **BI & Visualization Tools:** Microsoft Power BI (DAX Measures, Power Query ETL, Star Schema, KPI Dashboards), Tableau Desktop (Calculated Fields, LOD Expressions, Geographic Mapping)
- **Spreadsheets & Statistics:** Advanced Excel (Multi-Table Pivot Tables, Slicers, XLOOKUP, Nested IFS/COUNTIFS, Variance Modeling), Hypothesis Testing (Two-Sample t-Tests, ANOVA, Chi-Square, Pearson r)

FEATURED TECHNICAL PROJECTS

Retail E-Commerce Relational Database & Advanced SQL Analytics | *MySQL, Relational Schema Design (3NF), Window Functions, Joins*

- **Schema Architecture & Integrity:** Designed and implemented a normalized 6-table relational database (customers, products, orders, order_items, payments, reviews) with primary/foreign keys, CHECK constraints, and automated timestamps.
- **Advanced Query Analytics:** Authored a 6-tier analytical query suite utilizing multi-table INNER/LEFT/RIGHT JOINS, correlated subqueries (EXISTS/NOT EXISTS), and window functions (ROW_NUMBER() OVER PARTITION) for customer order ranking and gateway settlement reconciliation.

AegisLife Insurance: Risk Analytics & Claims Intelligence (Capstone) | *Python (Pandas, SciPy), SQLite, SQL DDL, Power BI*

- **Relational ETL & Statistical Rigor:** Ingested 2M+ records into normalized SQLite schema; ran parametric hypothesis tests in SciPy disproving legacy Risk Score predictive power ($r = 0.012$, $p = 0.64$) to recommend an underwriting model overhaul.
- **Fraud Detection & Power BI:** Isolated intermediary agent fraud clusters ($\chi^2 = 38.42$, $p < 0.001$) and delivered an interactive Power BI dashboard with C-suite briefing decks to mitigate a 49% claims suspicion rate.

Customer Lifetime Value & RFM Segmentation Analytics Pipeline | *Python (Pandas, NumPy, Seaborn), Jupyter, Pareto Decile Modeling*

- **RFM Quantile & Cohort Modeling:** Processed 23,050 orders across 1,000 accounts into 8 behavioral cohorts via pandas qcut quintile scoring; classified Champions (\$3.73M/16.2% rev) and Loyalists (\$5.12M/22.2% rev) for targeted lifecycle marketing.
- **Dual-Axis Pareto Revenue Analysis:** Built a dual-axis Pareto distribution chart mapping decile spend against cumulative revenue share, confirming top 20% of spenders generate \$6.23M (27.0%) and top 70% generate 78.3% of total revenue.

Enterprise Business Intelligence & Advanced Analytics Dashboards | *Tableau Desktop, Power BI (DAX), Advanced Excel*

- **Tableau Real Estate Analytics:** Analyzed 12 years of property transactions, mapping \$9.6B+ sales in Hot Springs and calculating Assessed-to-Sale ratios (1.08-1.11 benchmarks) for institutional discount valuation.
- **Power BI & Excel Modeling:** Engineered an Airline delay profitability dashboard in Power BI (quantifying delays >25 mins cause 18.4% profit erosion) and an Amazon multi-table sales quota variance model in Excel.

CERTIFICATIONS

- **IBM Data Analytics Certificate:** SQL & Databases, Python, Statistics, Excel, Power BI, Tableau, Data Transformation, Generative AI.

EDUCATION

- **Bachelor of Science (B.Sc.) in Physics, Chemistry, and Mathematics** | Kumaun University, Uttarakhand, India (Graduation: 2020)